Python Activity-Based Learning
Case Study

57 Smart Canteen Order and Stock Management

Selected case study | Prepared using the topics listed in the supplied assignment. ## 1. Problem

Statement

Create a simple canteen program that accepts customer orders, calculates bills, updates stock, rejects unavailable quantities, and displays remaining stock and daily sales.

2. Objectives

• Represent menu items and stock using lists.

• Process multiple orders with loops.

• Check stock before accepting an order.

• Calculate bills and update daily sales.

3. Assumptions

• The menu contains three items with fixed prices and opening stock.

• The user enters a menu number and quantity for each order.

• An order is accepted only if the quantity is positive and available.

• The program ends when the user enters 0 as the menu choice.

4. Algorithm

1. Create menu item, price, and stock lists.

2. Initialize total sales to zero.

3. Display the menu and read a choice.

4. If choice is 0, stop ordering.

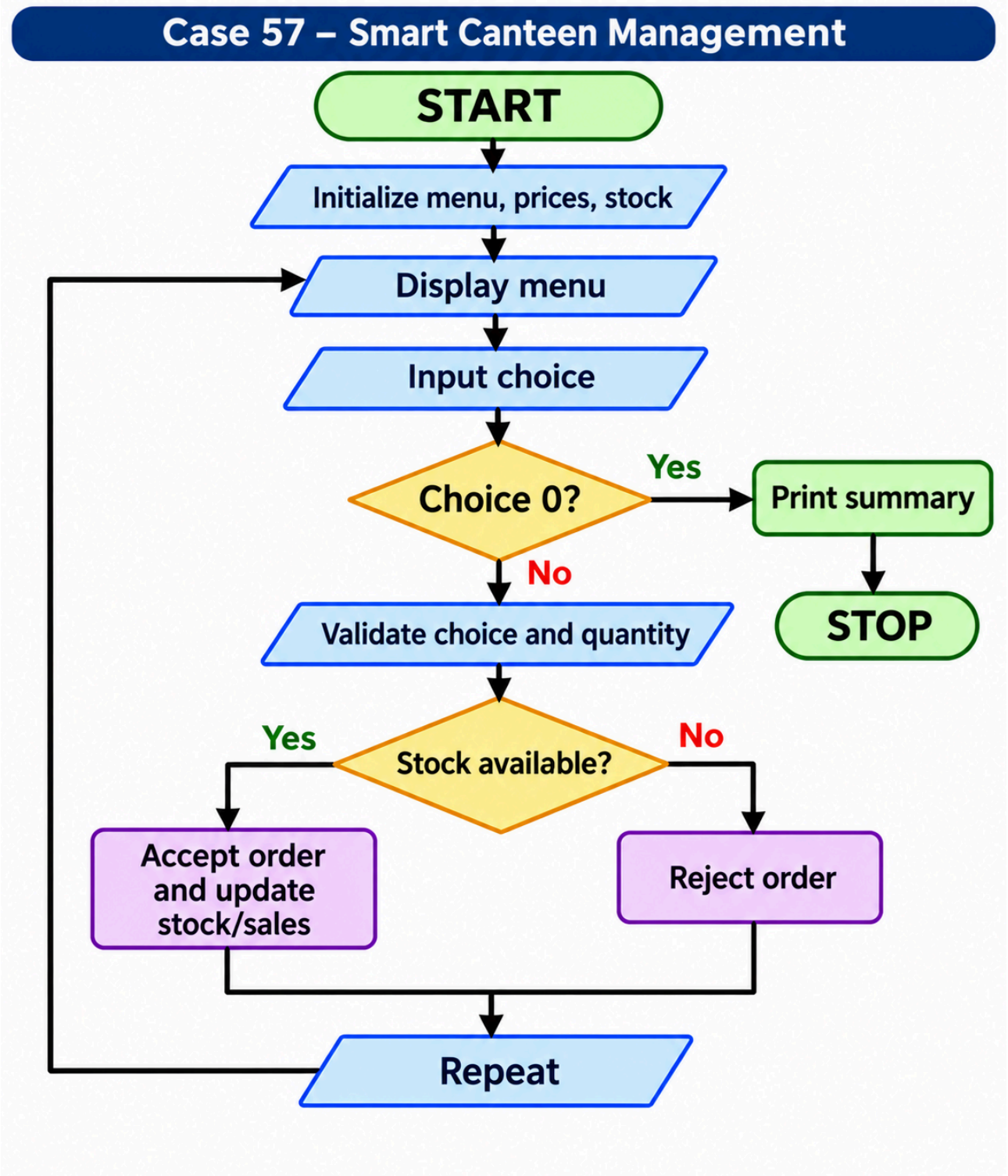
5. Validate the menu choice and quantity.

6. If stock is sufficient, calculate the bill, reduce stock, and update sales.

7. Otherwise reject the order.

8. At the end, display total sales and remaining stock.

5. Flowchart



6. Pseudocode

```
BEGIN
 menu, prices, stock ← predefined lists
 sales ← 0
 WHILE TRUE
 DISPLAY menu
 INPUT choice
 IF choice = 0 THEN BREAK
 IF choice is invalid THEN CONTINUE
 INPUT quantity
 IF quantity > 0 AND quantity <= stock[choice-1] THEN
 bill ← quantity × prices[choice-1]
 stock[choice-1] -= quantity
 sales += bill
 PRINT bill
 ELSE PRINT order rejected
```

```
 END IF
 END WHILE
 PRINT sales and remaining stock
END
```

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7. Python Source Code

```
menu = ["Idli", "Dosa", "Tea"]
prices = [30, 50, 10]
stock = [20, 15, 40]
total_sales = 0

while True:
 print("\n--- CANTEEN MENU ---")
 for i in range(3):
 print(i + 1, menu[i], "- Rs.", prices[i],
 "| Available:", stock[i])
 print("0 Exit")

 choice = int(input("Choose item number: "))

 if choice == 0:
 break
 elif choice < 1 or choice > 3:
 print("Invalid menu choice")
 continue

 quantity = int(input("Enter quantity: "))
 index = choice - 1

 if quantity <= 0:
 print("Quantity must be positive")
 elif quantity > stock[index]:
 print("Order rejected: insufficient stock")
 else:
 bill = quantity * prices[index]
 stock[index] -= quantity
 total_sales += bill
 print("Order accepted. Bill: Rs.", bill)

print("\n--- DAILY SUMMARY ---")
print("Total sales: Rs.", total_sales)
for i in range(3):
 print(menu[i], "remaining stock:", stock[i])
```

8. Test Cases

Input / Scenario	Expected result
Order 2 Dosa; then exit	Bill ■100; Dosa stock becomes 13; sales ■100
Order 50 Tea	Rejected: insufficient stock; stock unchanged
Enter menu choice 0	Program exits and displays summary

9. Conclusion

The program combines lists, loops, conditions, arithmetic, and stock updates to simulate a basic canteen ordering system.

Note: Run the program in Python 3. The sample test cases are illustrative; enter the values when prompted.